

Review retrospective:
*the affine-bilinear covariance specialisation — a clean
composition of the Gibbs algebra*

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Abstract

A soundness review of `Laplace/AffineBilinearCov.lean`: the affine-bilinear specialisation of the Gibbs covariance, $\text{Cov}_t[a_1\varphi + b_1, a_2\psi + b_2] = a_1a_2 \text{Cov}_t[\varphi, \psi]$, the first consumer of the covariance algebra from `Laplace.Gibbs`. Result: a clean fidelity pass with a single **dead-open** removal.

1 Setting

The immediate downstream of the Gibbs foundation: it composes `gibbsCov_add_left/right`, `gibbsCov_smul_left/right`, and `gibbsCov_const_left/right` into the affine-input reduction — additive constants drop (covariance with a constant is zero), scalars factor out.

2 What was reviewed

`gibbsCov_affine_left` ($\text{Cov}[a\varphi + b, \psi] = a \text{Cov}[\varphi, \psi]$), `gibbsCov_affine_right` (its symmetric twin), and the headline `gibbsCov_affine_affine` ($a_1a_2 \text{Cov}$), checked against the affine-bilinear law and for correct, minimal integrability hypotheses.

3 Fidelity / soundness findings

Sound; no defect. The reviewer confirmed all three statements: the headline coefficient is the correct product a_1a_2 , both additive constants disappear, and the four base integrability hypotheses ($h_{\text{exp}}, h_{\varphi}, h_{\psi}, h_{\varphi\psi}$) are exactly the compositional ones needed to derive integrability of the scaled/offset observables before applying the base algebra — all used, none spurious. “No statement-level mismatches found.”

4 Simplifications applied

One dead open. `open Real` carried nothing (every `Real.exp` is qualified; the only bare `exp` is in the module docstring); `open MeasureTheory` stays (bare `Integrable` in every hypothesis), and `import Laplace.Gibbs` is the load-bearing source of the algebra. Build green, no warnings, signatures byte-identical; the reviewer affirmed. No dead locals and no droppable hypotheses — a tight composition file.

5 Disagreements and open questions

None. The reviewer’s proof-golf (inline the syntactic `eq/change` rewrites; prove `affine_right` directly from the right-slot algebra to drop a derived local) was declined as proof-shape-only.

6 Lessons

- **A pure composition file inherits its parent’s cleanliness.** Having just reviewed the Gibbs algebra it consumes, this file’s review reduced to confirming the composition is faithful and the hypotheses minimal — exactly the cheap, high-confidence pass that reviewing a consumer right after its foundation buys (the same economy seen lifting the TwoD trio right after their `AddSeparable` root).
- **The dead open `Real` recurs once more.** Across the laplace seabed the single most common landable is a templated open `Real` that the file never uses unqualified; the foundational and composition files (`Gibbs`, `AffineBilinearCov`) are no exception.

7 Follow-ups

- Remaining never-reviewed laplace files: the root `AffineKappa3` (310L) and the large `Multi/` analytic files (`RescaledIntegrals` 1547L, and the `Covariance/CovarianceSharp/CovarianceExplicit` trio up to 19.6k lines, for sectioned review).